ES 1402 Final Presentation

Group Primitive: Hudson Mills, Kyuri Lim, Youssef Nassar, Gümüşsu Memik

Some background

Notes and pitch

$$f = 440 \cdot 2^{(n-49)/12}$$

Pitch is frequency. Each piano key n raises f by a factor of 1.0595, each “octave" doubles frequency

Bottle (Helmholtz resonator)

$$f = \frac{c}{2\pi}\sqrt{\frac{A}{VL_{\text{eff}}}}$$

Air in the neck is the mass, air in the bottle is the spring. More air gives a lower pitch ($f \propto 1/\sqrt{V}$).

Straw flutes (tubes)

$$f_{\text{closed}} = \frac{c}{4L} \quad f_{\text{open}} = \frac{c}{2L}$$

Closed at one end: odd harmonics, L is a quarter wavelength. Open at both ends: L is a half wavelength. Longer tube, lower pitch.

Tuned Pipes

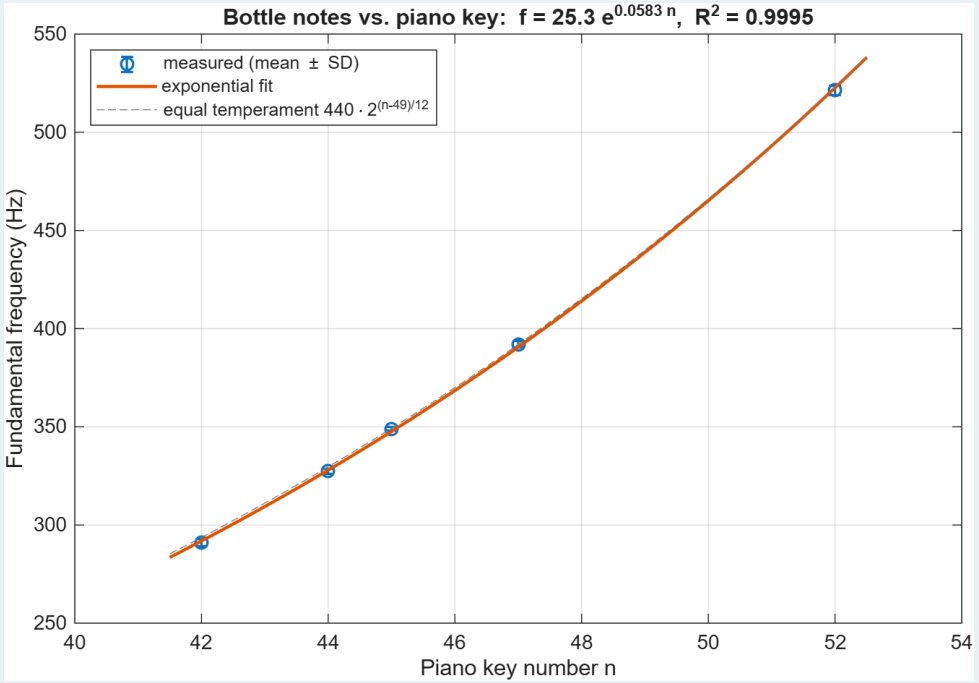
$$f \propto \frac{t}{L^2}$$

A pipe free at both ends vibrates faster when it is thicker (t) and shorter (L). Cutting a pipe shorter raises its pitch.

Water-bottle (“Helmholtz resonators”) notes



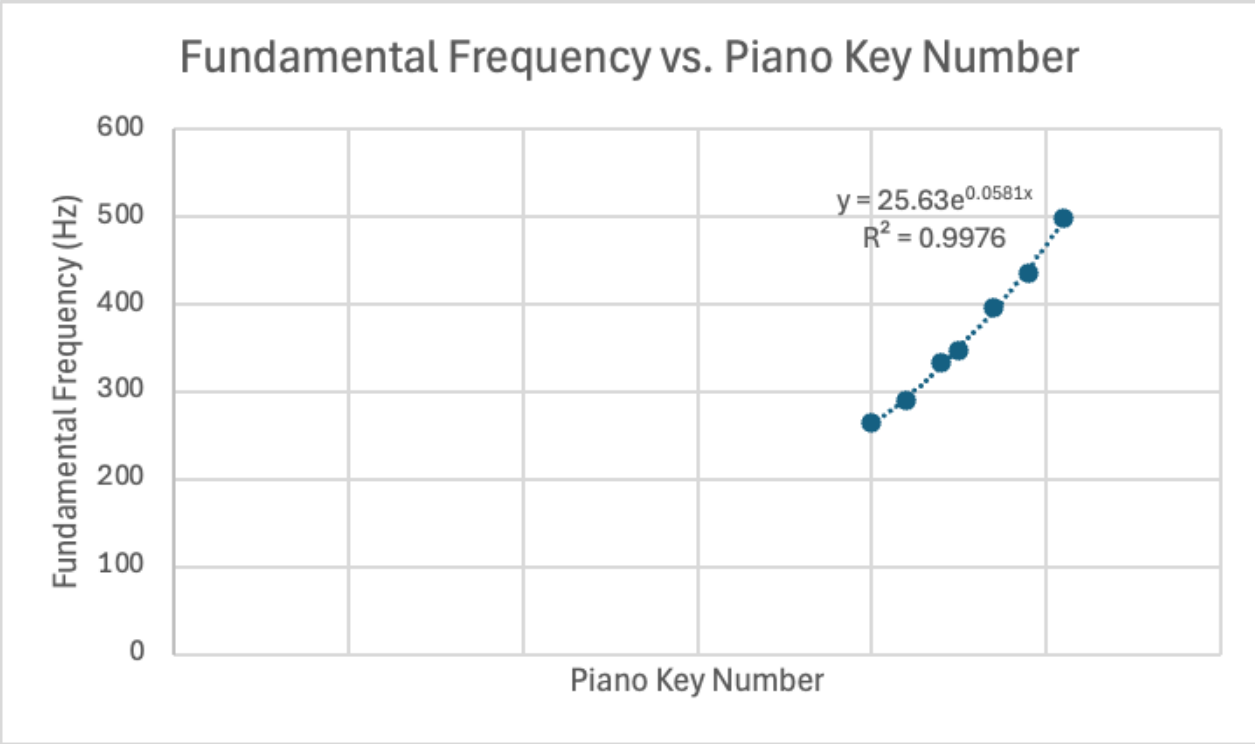
Bottles (Youssef): 5 identical 355 mL bottles, each tuned by pouring water out with a syringe until Friture showed the target note: D4, E4, F4, G4, C5.



$$f = 25.3\, e^{0.0583\, n}, \qquad R^2 = 0.9995$$

All five notes within 1 % of the piano frequencies.

Closed-end straw flutes



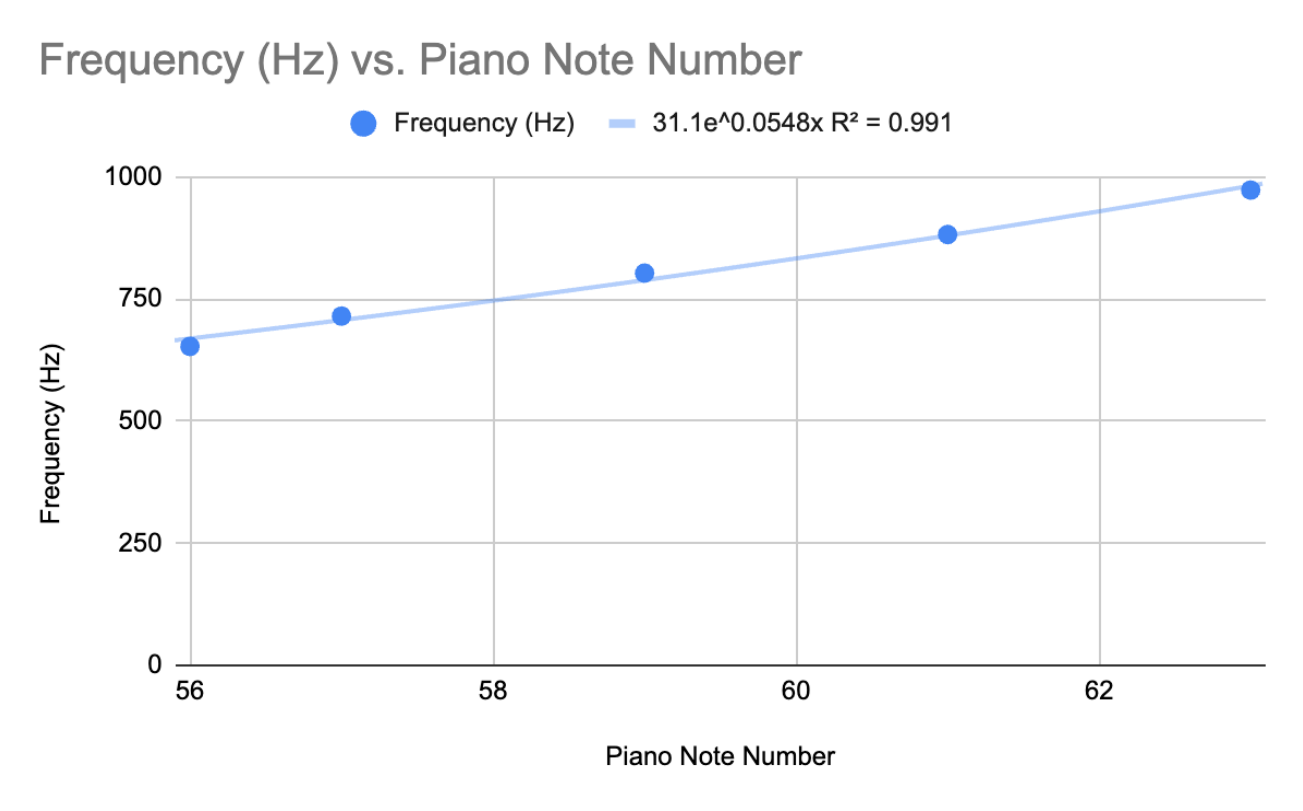


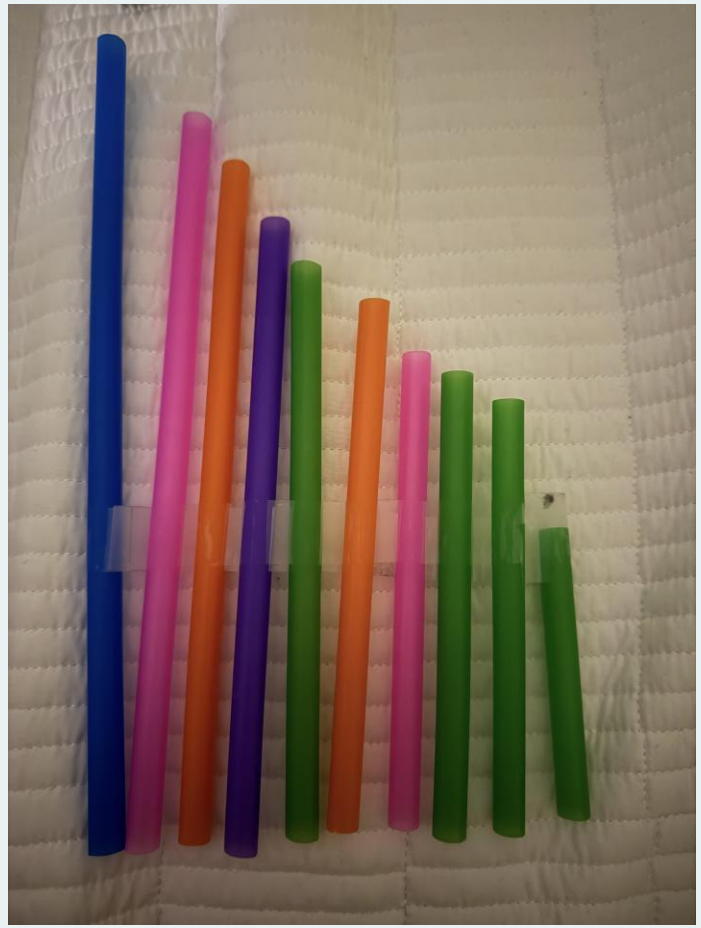
Gumussu: straws closed at one end, each blown across the top. Tuned by length, f = c/4L .
Notes: A4, B4, C4, D4, E4, F4,G4.

$$f_{\text{closed}} = \frac{c}{4L}$$

$R^2 = 0.9976$

Open-end straw flutes



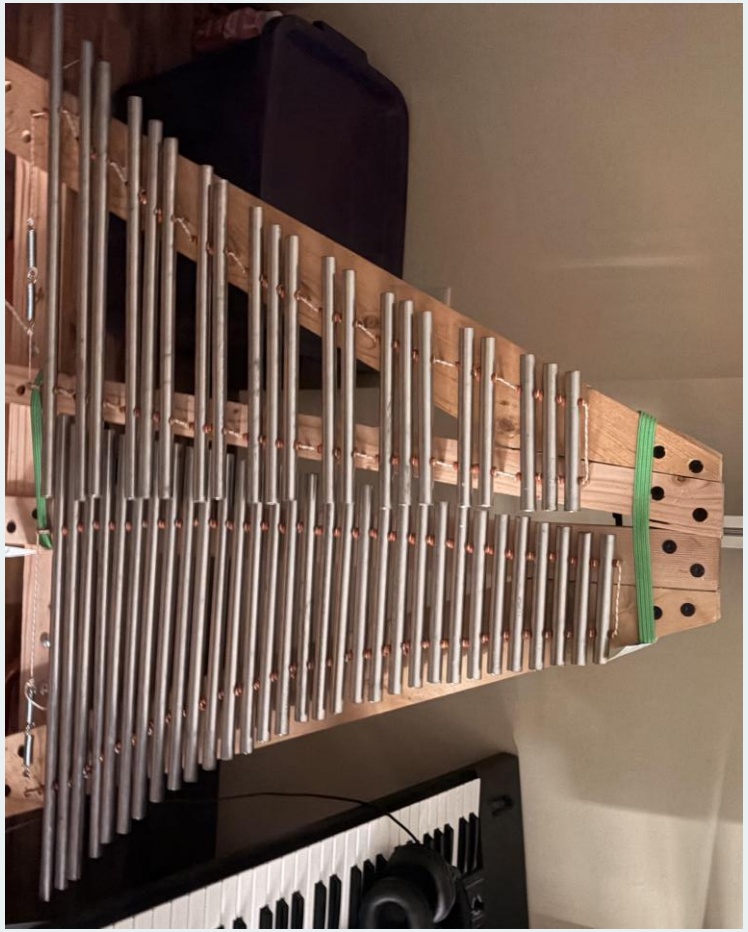


Kyuri: straws open at both ends, each blown across the top. Tuned by length, f = c/2L (plus an end correction). Notes: E5, F5, G5, A5, B5

Fit: 31.1e^0.0548x, $R^2 = 0.991$

$$f_{\text{open}} = \frac{c}{2L}$$

Full-range Gamelan

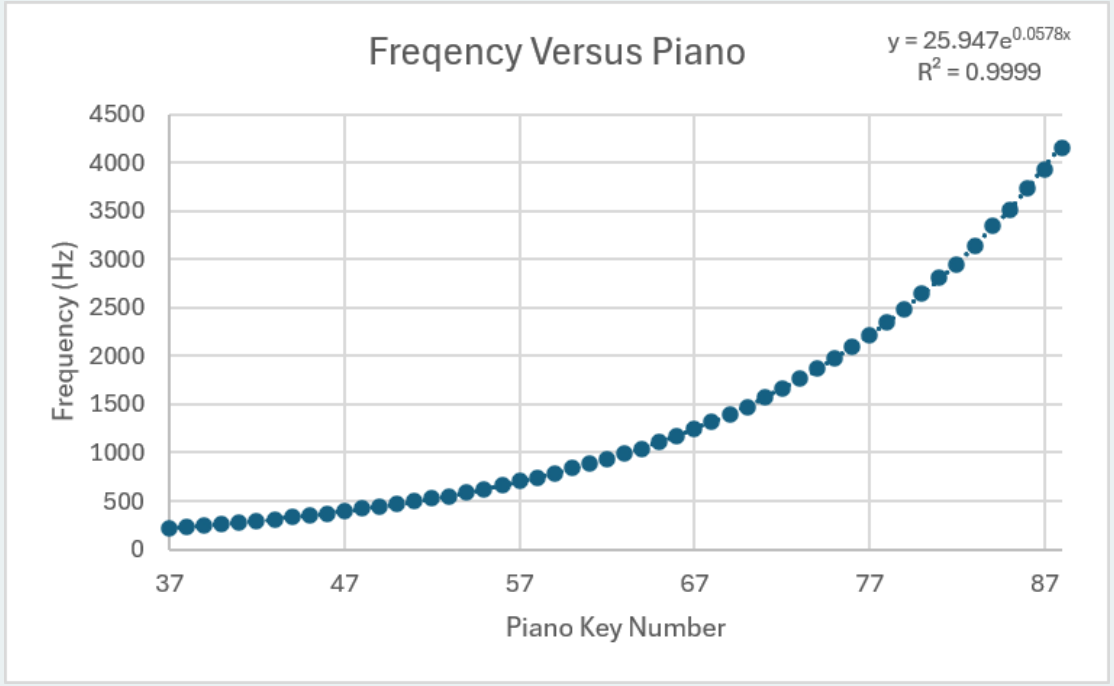


$$f \propto \frac{t}{L^2}$$

Hudson: 52 bars covering A3 to C8, the widest range of our group.

Bars: ¾" Stainless Steel. Tuned by cutting each bar to length: $f \propto t/L^2$.

Checked each note against the piano frequency in Friture.



Results and lessons

Went well

- Every instrument plays a scale in the same piano tuning
- Bottle notes within 1% of the piano, $R^2 = 0.9995$
- Different instruments used different physics: bottle, closed tube, open tube and bars.
- The Gamelan covers the widest range of the four.

Did not go so well

- Friture peaks were wide and noisy, hard to read
- Background noise made frequency measurements difficult.
- Blowing at slightly different angles sometimes produced weak or unstable notes.

Learned working together

- Testing early found problems sooner.

Do they sound good together? We will let Hudson's overkill instrument decide!

